

Course project

Artificial Intelligence

Deadline: 9th May 2024

Instructions:

1. **It is a group assignment of three members.**
2. **Each group has to select one project from the following two projects.**
3. **For both projects, students must select unique self-created datasets and must have to mention the dataset description in the provided google sheet.**
4. The assignment should be implemented using the Python programming language.
5. Strict adherence to academic integrity is required, with no plagiarism permitted. Any form of copying, whether from other students or online sources, will result in zero marks for the entire assignment.
6. The code must be well-commented for clarity and understanding.
7. Submissions must be made on time, as late submissions will be penalized.
8. Incomplete submissions lacking either the source code or the accompanying report will not receive any marks. The report should provide a concise explanation of the steps involved in the program for each question, along with the obtained results.

To submit the assignment, compress both the source code and the report into a single folder, named as `ROLLNO_NAME.ZIP`

Project: 01

The demand for fine-tuned models for chatbots arises from the need for tailored and specialized assistance across various domains. As individuals navigate through specific fields such as medicine, law, engineering, or finance, they encounter complexities that demand precise and accurate information. For instance, consider a medical graduate undertaking a house job. Despite their foundational knowledge, the vast array of medical terminology and procedures can be overwhelming. In such scenarios, having access to a chatbot fine-tuned on medical literature becomes invaluable. This specialized chatbot can serve as a reliable assistant, providing authentic terms, procedures, and guidelines directly from medical textbooks. The motivation for domain-specific chatbots lies in their ability to cater to the unique needs of professionals, students, and enthusiasts within specific domains, offering targeted assistance and enhancing productivity and knowledge retention.

In this assignment, students are tasked with developing a domain-specific chatbot by fine-tuning a pre-existing model on a specified book relevant to the chosen domain. For example, students may choose to fine-tune a BERT model on a medical textbook and name the resulting chatbot "MedBot." The objective is to train the chatbot to understand and respond to user queries pertaining to the content of the book, providing accurate and relevant information.

To accomplish this, students will first select a suitable pre-trained language model and acquire a dataset consisting of the text from the chosen book. They will then fine-tune the model using techniques such as transfer learning, adjusting the model's parameters to better understand the domain-specific language and concepts present in the book.

In addition to training the model, students will design and implement a simple graphical user interface (GUI) for the chatbot. The GUI should allow users to input queries or prompts and receive responses from the chatbot in a user-friendly manner. This interface enhances the accessibility and usability of the chatbot, making it more intuitive for users to interact with.

Throughout the assignment, students will gain hands-on experience with natural language processing techniques, including fine-tuning pre-trained models, dataset preprocessing, and GUI development. They will also explore the practical applications of domain-specific chatbots in providing targeted assistance and information retrieval within specialized fields.

Requirements:

1. Dataset
2. Code files
3. GUI

Project 2:

The motivation behind this project stems from the increasing demand for automated billing systems in retail environments and the potential of artificial intelligence (AI) to revolutionize traditional processes. By integrating AI techniques such as object detection and image recognition, this project aims to develop an automated billing system capable of accurately identifying products from images and generating total bills based on their prices. The objective is to streamline the billing process, reduce errors, and enhance customer satisfaction by providing a fast and efficient checkout experience.

For this project, each group will select a unique dataset containing images of products from various categories, such as crockery, makeup, or electronics, along with their corresponding prices. Utilizing object detection models, students will train algorithms to recognize and classify individual products within the images. Once identified, the system will extract the relevant pricing information and calculate the total bill based on the detected products. The final step involves

presenting the total bill on a graphical user interface (GUI), allowing users to review and confirm their purchases seamlessly.

Through this project, students will gain practical experience in AI, computer vision, and GUI development, while also understanding the importance of automation in retail processes. By building an AI-powered billing system, students will contribute to the advancement of retail technology and acquire valuable skills applicable across various industries.

Requirements:

1. Dataset
2. Code files
3. GUI